SOFTWARE CHANGE

IMPLEMENTATION REPORT

CR-020: Student Change Password

Field	Details
Change Request ID	CR-020
Change Request Title	Student Change Password
Student Name	N. Bavanika
Student ID	JFICT24_19
Implementation Status	Completed
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

The Student Management System should allow students to change their own account passwords securely through the Student Dashboard. The system must verify the current password before accepting a new password and ensure that the new password and confirmation password match.

The main objectives are:

- Provide a Change Password option on the Student Dashboard.
- Allow students to enter their current password, new password, and confirmation password.
- Verify the current password before changing the account password.
- Validate the new password and ensure the confirmation matches.
- Store the new password securely as a hash in the database.
- Display appropriate success and error messages.
- Preserve existing authentication, authorization, and student management functionality.

1.2 Existing Behaviour

Before implementation:

- Students did not have a dedicated Change Password option on the Student Dashboard.
- Students could not use a dedicated form to change their account passwords.
- The application did not provide the requested self-service password-change workflow.
- A secure password-change process was required to ensure that students could update only their own passwords.

1.3 Requested Change

Add a Change Password feature to the Student Dashboard that allows authenticated students to update their passwords securely.

The implementation must:

- Provide Current Password, New Password, and Confirm New Password fields.

- Validate all required fields.
- Verify the current password using bcrypt.
- Validate the new password according to the configured password-strength rules.
- Ensure the new password and confirmation password match.
- Reject reuse of the current password.
- Hash the new password before updating the database.
- Restrict the endpoint to authenticated users with the STUDENT role.
- Preserve existing authentication, authorization, and dashboard functionality.

2. Implementation

2.1 Implementation Summary

CR-020 was implemented across the frontend, backend API, controller, and student model.

A Change Password option was added to the Student Dashboard. Selecting this option opens a modal form containing Current Password, New Password, and Confirm New Password fields.

The frontend validates the submitted fields and sends a PATCH request to /api/students/change-password through the existing API service.

The backend route is protected by authentication middleware and STUDENT role authorization. The controller validates the request, verifies the current password using bcrypt.compare(), checks the new password, hashes it using bcrypt, and updates the password hash in the database.

The student model provides the getPasswordHashByUserId() and updatePasswordHash() functions to retrieve and update the password hash in the users table.

The feature provides success and error feedback and supports clearing and cancelling the form. Existing authentication and student management functionality is preserved.

2.2 Affected Files

File / Component	Change	Purpose
frontend/src/pages/StudentDashboardPage.jsx	Modified	Add the Change Password option to the Student Dashboard.
frontend/src/components/ChangeStudentPassword.jsx	Added	Provide the password-change modal, input fields, validation, API request, and feedback messages.
frontend/src/services/api.js	Used	Send the authenticated API request through the shared API service.
frontend/src/styles/studentDashboard.css	Modified	Style the password-change modal, form controls, and responsive layout.
backend/src/routes/studentRoutes.js	Modified	Add the protected PATCH /change-password route.
backend/src/controllers/studentController.js	Modified	Validate password-change requests, verify the current password, hash the new password, and return the result.
backend/src/models/studentModel.js	Modified	Retrieve the existing password hash and update the password hash in the database.
Existing authentication and authorization	Preserved	Maintain the existing authentication middleware and role-based access control.

2.3 Technical Changes

A Change Password modal was added to the Student Dashboard.

The form contains the following fields:

- Current Password
- New Password
- Confirm New Password

The frontend validates the required fields and checks that the new password matches the confirmation password before submitting the request.

The request is sent using:

```
PATCH /api/students/change-password
```

The backend endpoint uses authentication middleware and STUDENT role authorization.

Password Verification and Update

The backend retrieves the authenticated student's password hash from the `users` table.

The current password is verified using:

```
bcrypt.compare(currentPassword, user.password_hash)
```

After successful verification and validation, the new password is hashed using bcrypt. The model updates the `users.password_hash` field with the new hash.

The implementation also validates the password confirmation and rejects invalid password-change requests.

Student Dashboard

The Change Password option opens the modal form. The frontend displays appropriate success or error messages based on the API response.

The interface also provides Clear and Cancel actions and supports a responsive layout.

2.4 Implementation Logic

- The student opens the Student Dashboard.
- The student selects the Change Password option.
- The system displays the password-change modal.
- The student enters the current password, new password, and confirmation password.
- The frontend validates the required fields.
- The system checks whether the new password and confirmation password match.
- The frontend sends a PATCH request to `/api/students/change-password`.
- The backend authenticates the request and verifies the STUDENT role.

- The controller identifies the student using the authenticated user ID.
- The system retrieves the current password hash from the database.
- The current password is verified using bcrypt.
- The system validates the new password and rejects reuse of the current password.
- The new password is hashed using bcrypt.
- The model updates the password hash in the users table.
- The frontend displays the successful password-change message.
- The student can use the new password for subsequent login.
- Existing authentication, authorization, and Student Dashboard functionality continue to work.

2.5 Error and Data Handling

The implementation preserves the existing authentication and authorization process and includes validation for password-change requests.

For Password Change:

- Required fields are validated before submission.
- An incorrect current password is rejected.
- A new password that does not meet the configured password-strength rules is rejected.
- A mismatched confirmation password is rejected.
- Reuse of the current password is rejected.
- Only authenticated users with the STUDENT role can access the endpoint.
- The new password is hashed before being stored in the database.

For Database and UI Handling:

- The current password is compared with the stored hash using bcrypt.
- The authenticated user's ID is used to identify the account.
- The database is updated only after successful validation.
- The frontend displays appropriate success and error messages.
- Passwords are not stored as plain text.
- Existing authentication and student management functionality is preserved.

3. Testing & Verification

3.1 Testing Approach

- Testing Environment: Development
- Browser: Google Chrome
- Test Role: Student
- Test Date: 13 October 2026

The following testing activities were performed:

- Password-change form and modal testing.
- Required-field validation testing.
- Current-password verification testing.
- New-password strength validation testing.
- Confirmation password matching testing.

- Password reuse rejection testing.
- Successful password update testing.
- Database password hash verification.
- Old-password and new-password login verification.
- Authentication and authorization testing.
- Error handling testing.
- User interface testing.
- Existing Student Dashboard regression testing.

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open Change Password form	The password-change form opens correctly.	The form opened correctly.	Pass
TC-02	Submit empty fields	Required-field validation is displayed.	Required-field validation worked correctly.	Pass
TC-03	Enter incorrect current password	The request is rejected and the password remains unchanged.	The incorrect current password was rejected.	Pass
TC-04	Enter a weak new password	Password-strength validation is displayed.	Password-strength validation worked correctly.	Pass
TC-05	Enter mismatched new password and confirmation	A confirmation mismatch message is displayed.	Confirmation validation worked correctly.	Pass
TC-06	Reuse the current password	Password reuse is rejected.	Password reuse was rejected.	Pass
TC-07	Submit valid password details	The password is updated and a success message is displayed.	The password was changed successfully.	Pass
TC-08	Log in using the old password	The old password is rejected after the change.	The old password was rejected.	Pass
TC-09	Log in using the new password	The new password is accepted.	The new password was accepted.	Pass
TC-10	Use the Clear action	All password fields are cleared.	The fields were cleared correctly.	Pass
TC-11	Cancel or close the modal	The modal closes without submitting a change.	The modal closed correctly.	Pass
TC-12	Access endpoint without authentication	The request is rejected.	The unauthenticated request was rejected.	Pass
TC-13	Access endpoint using a non-student role	The request is rejected by role authorization.	Role authorization worked correctly.	Pass
TC-14	Verify database after password change	The password hash is updated and remains hashed.	The password hash was updated correctly.	Pass
TC-15	Test responsive UI and existing dashboard features	The interface works correctly and existing features remain functional.	The interface and existing dashboard features worked correctly.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	15
Passed	15
Failed	0
Pending / Not Verified	0
Pass Rate	100% Pass

4. Acceptance Criteria Verification

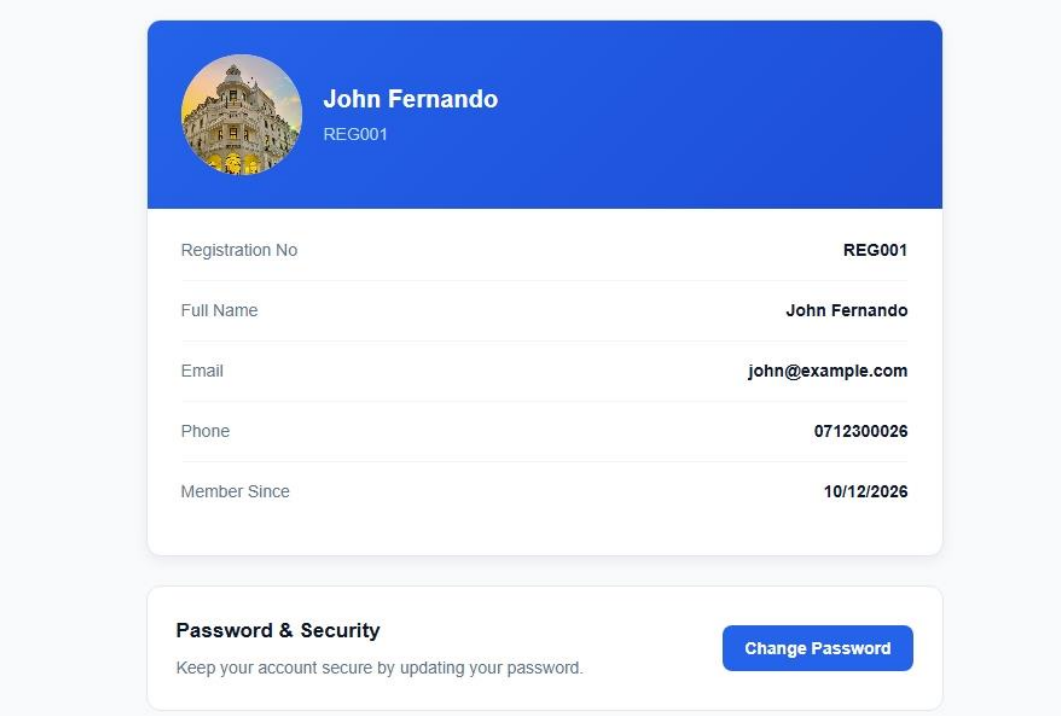
ID	Acceptance Criterion	Verification	Status
AC-001	Students can open the Change Password form from the Student Dashboard.	Student Dashboard UI verification.	Pass
AC-002	The system verifies the current password before updating the password.	Current-password verification testing.	Pass
AC-003	The new password must meet the configured rules and match the confirmation.	Password validation testing.	Pass
AC-004	The new password is stored as a secure hash.	Database verification.	Pass
AC-005	Only authenticated students can use the password-change endpoint.	Authentication and role authorization testing.	Pass
AC-006	The new password works and the old password no longer works.	Login verification after password change.	Pass
AC-007	The interface displays appropriate success and error messages.	Frontend and API response testing.	Pass
AC-008	Existing authentication and dashboard functionality remain intact.	Regression testing.	Pass

5. Evidence

Evidence 01 – Student Change Password Form

Figure 1: Student Dashboard displaying the Change Password option and the password-change modal with Current Password, New Password, and Confirm New Password fields.

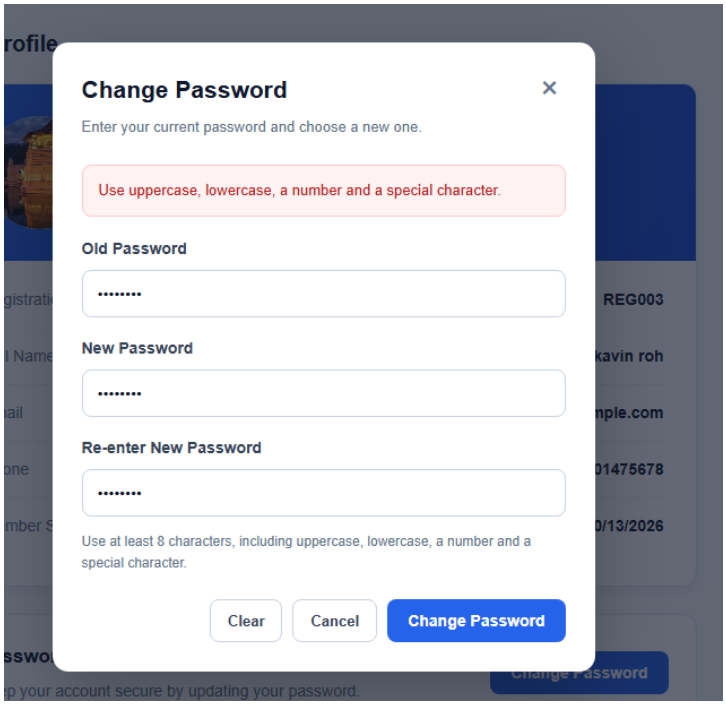
Related acceptance criteria: AC-001, AC-003.



Evidence 02 – Password Validation

Figure 2: Password-change form displaying validation messages for an incorrect current password, mismatched confirmation, or invalid new password.

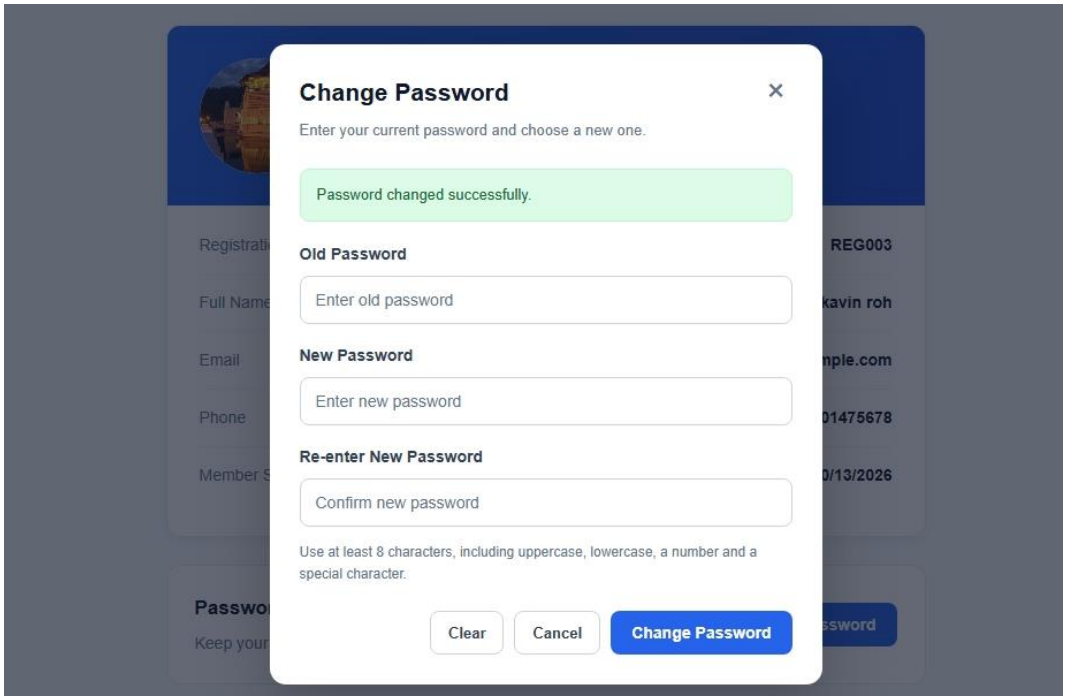
Related acceptance criteria: AC-002, AC-003.



Evidence 03 – Successful Password Change

Figure 3: Password-change form displaying the success message after the student successfully changes their password.

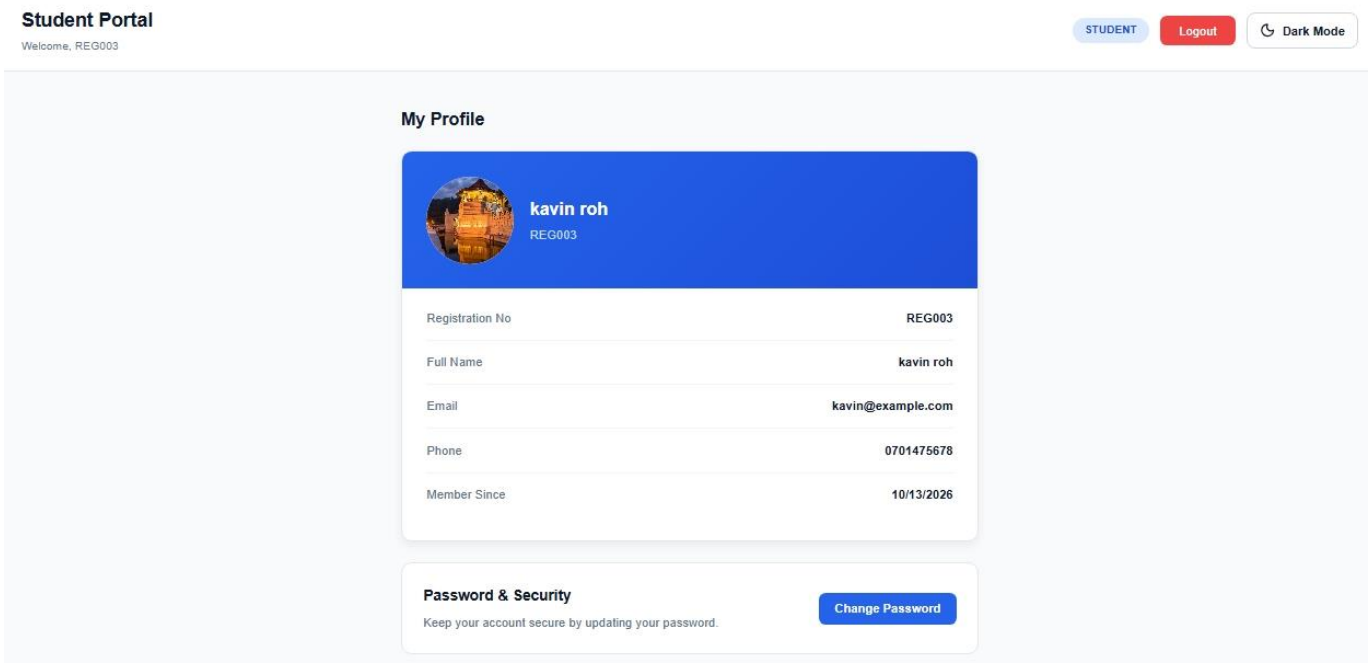
Related acceptance criteria: AC-004, AC-006, AC-007.



Evidence 04 – Login Verification

Figure 4: Login verification demonstrating that the new password is accepted and the old password is rejected after a successful password change.

Related acceptance criteria: AC-005, AC-006.



6. Final Result

Item	Result
Requirements	Completed
Student Dashboard Change Password Option	Completed
Password-Change Modal	Completed
Required-Field Validation	Completed
Current Password Verification	Completed
New Password Validation	Completed
Confirmation Password Validation	Completed
Secure Password Hash Update	Completed
Backend Password-Change Endpoint	Completed
Authentication and Role Authorization	Completed
Success and Error Feedback	Completed
Testing	15 of 15 Passed
Acceptance Criteria	All Met
Existing Authentication	Verified
Existing Student Dashboard Functionality	Verified
Final Status	Completed

6.1 Final Summary

CR-020 was successfully implemented to allow students to change their account passwords securely through the Student Dashboard.

The new feature provides a password-change modal with Current Password, New Password, and Confirm New Password fields. The frontend validates the submitted information and communicates with the backend through the existing API service.

The backend endpoint is protected by authentication and STUDENT role authorization. The current password is verified using bcrypt before the new password is hashed and stored in the database. The system validates password strength, checks the confirmation password, and rejects invalid password-change requests.

The implementation preserves the existing authentication process and Student Dashboard functionality. The interface provides appropriate feedback and supports clearing and cancelling the form.

A total of 15 test cases were completed, with all 15 passing and no failures. All defined acceptance criteria were verified and marked as met.

The change improves account security by providing students with a secure self-service password-management feature.

6.2 Issues / Limitations

- The feature changes the current password and does not introduce a separate password-history feature.
- The student must know the current password to change the account password through this workflow.
- The new password is stored as a hash rather than plain text.
- The endpoint is protected by authentication and STUDENT role authorization.
- No outstanding issues were identified for the requested Student Change Password feature.

Student Name: N. Bavanika
Student ID: JFICT24_19
Date: 13 October 2026
Signature:

~ END OF REPORT ~